

Cross-View Action Consistency for Camera-Robust Vision-Language-Action Policies

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Abstract: Vision-language-action (VLA) policies fine-tuned from a fixed scene camera can fail when the camera is moved, even when the task, objects, language, and robot state are unchanged. We study scene-camera viewpoint robustness using only a scene RGB image, language, and proprioception, without camera labels, extrinsics, depth, or point-cloud inputs. The wrist stream is masked throughout to prevent an unperturbed visual shortcut from confounding attribution to scene-camera variation. For flow-based VLAs, we propose to regularize the action-flow velocity field, the quantity directly integrated to generate continuous action chunks. We construct action-equivalent view pairs by resetting original LIBERO demonstrations to the same MuJoCo state and rendering nominal and perturbed scene-camera views. Both views are supervised by flow matching, while a cross-view loss encourages their predicted action-flow velocities to agree at the same sampled flow coordinates. On the LIBERO-Plus camera-perturbation track, our method reaches $87.2 \pm 0.4\%$ (4,797 rollouts per seed across 3 training seeds), +7.4pp over flow-matching-only training on the same paired data ($79.8 \pm 0.8\%$, also 3 seeds) and +12.5pp over naive mixed-camera SFT, while maintaining nominal-camera ID performance ($95.0 \pm 0.8\%$; same-data FM-only: $95.0 \pm 4.3\%$). A shuffled-pair control collapses to 25.8%, showing that the gain depends on action-equivalent pairing. On a real robot, we evaluate three tabletop tasks with 10 rollouts per task and camera placement; held-out-camera success improves from 53.3% to 74.4% under the same single-scene-RGB inference interface. Project page: <https://anonymous0311.github.io/pages-cross-view-action-consistency-vla>.

Keywords: Vision-Language-Action Models, Camera Robustness for VLAs, Flow Matching, Multi-View Supervision

1 Introduction

Vision-language-action policies map RGB observations, language instructions, and robot state to robot actions [1, 2, 3, 4, 5]. In practice, a pretrained VLA is often adapted from demonstrations collected with a fixed scene camera, but the same policy may degrade when the camera is bumped, re-mounted, or moved in distance, height, or orientation. Existing remedies often change the sensing or inference contract through depth, point clouds, calibrated multi-view inputs, camera rays, camera labels, novel-view rendering, or geometry-aware architectures [6, 7, 8, 9, 10, 11, 12, 13, 14]. Camera-diverse supervised fine-tuning is simpler, but data diversity alone does not impose action-level invariance between equivalent views. We ask whether a flow-based VLA can become robust to scene-camera motion without depth, point clouds, camera labels, or geometric inputs.

The desired invariance is action-level. If different images observe the same physical state and share the same instruction and robot state, the policy should intend the same action. For flow-based VLAs, actions are generated by integrating a learned velocity field $v_\theta(x_t, t \mid o, l, q)$ over noisy

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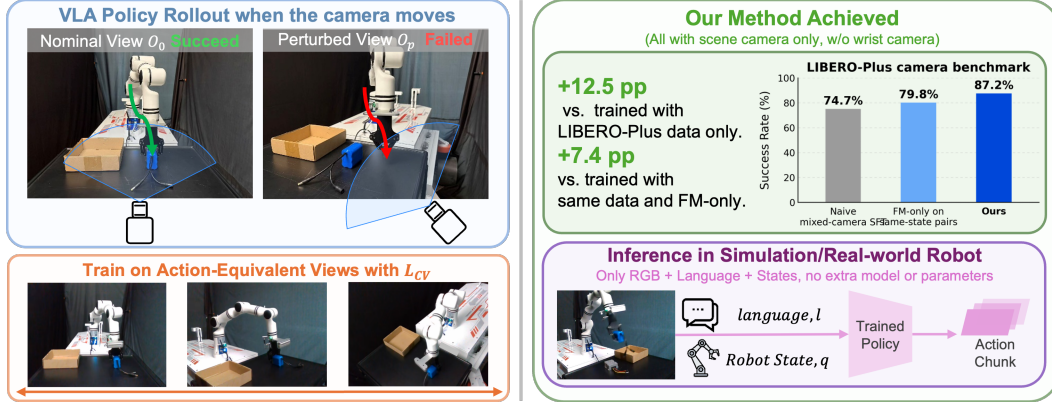


Figure 1: Camera viewpoint shifts can cause a VLA policy to produce different actions from the same physical state. We train on action-equivalent views so that different views of the same state are supervised by the same demonstrated action and regularized to produce consistent action-flow predictions. At inference, the policy receives a single RGB scene image (held-out camera placement), language instruction, and proprioceptive state; the wrist stream is masked throughout as an experimental control.

action chunks. We therefore regularize the object that directly produces actions: the local action-flow velocity prediction at the same flow coordinate. This differs from view-invariant representation learning, where matching an intermediate feature does not necessarily imply matching the action distribution.

We implement this idea with action-equivalent multi-view pairs: two scene images of the same physical state (nominal and perturbed camera) sharing language, proprioception, and demonstrated actions. During training, both views receive flow matching to the same action target, and their predicted velocities are encouraged to agree at the same sampled flow coordinate; this cross-view loss is used only during training.

We evaluate on the LIBERO-Plus camera-perturbation track [15, 16] against a nominal-only baseline, camera-diverse supervised fine-tuning, and—most informatively—flow-matching-only training on the same paired data, the last of which isolates the cross-view term from pair-data exposure. The proposed objective reaches $87.2 \pm 0.4\%$ over three training seeds, $+7.4\text{pp}$ over the same-data FM-only control, which also uses the same three seeds. A shuffled-pair control that breaks action equivalence inside the cross-view loss while preserving per-view flow-matching labels collapses performance, showing that the gain depends on action-equivalent correspondence rather than generic smoothing. A real-robot held-out-camera study with synchronized multi-camera demonstrations confirms that the same training principle transfers beyond simulator rerendering.

Contributions. (1) We formulate scene-camera viewpoint robustness for flow-based VLAs as local action-flow consistency across action-equivalent views, and propose a cross-view action-flow consistency objective that regularizes the velocity field for action generation directly, while adding no sensors, calibration, or camera-specific inputs at inference. (2) Controlled ablations show that matched cross-view action-flow consistency improves over camera-diverse flow matching, and that its benefit depends on action-equivalent pairing rather than generic regularization.

2 Related Work

Vision-language-action policies and flow action heads. VLAs combine visual encoders, language-conditioned transformers, and action decoders to control robots from RGB observations and instructions [1, 2, 3]. Recent systems generate continuous action chunks with diffusion or flow-style objectives [17, 4, 5, 18]. We build on this formulation: the policy predicts a velocity field in action space, and our regularizer acts directly on that field.

Camera and geometry-aware robot policies. A large body of work improves viewpoint robustness by changing the deployment-time inputs or architecture: view synthesis [6], spatial or geometry-aware VLA architectures [7, 8], point-cloud and 3D-aligned inputs [9, 10], camera-conditioned policies [11, 13], orthographic view generation [14], observation-centric action formulations [12], and lightweight learned recalibration of the visual representation [19]. These remedies modify the inference contract; ours acts only on the training objective and leaves the single-scene-RGB, calibration-free interface unchanged, so it is largely orthogonal to and combinable with them. Appendix H contrasts these inference-contract requirements across method families.

Multi-view supervision and invariance. Multi-view data can teach representations or policies to ignore nuisance viewpoint changes [20, 21]. Generic robot visual pretraining also improves downstream manipulation [22, 23, 24, 25]. However, feature invariance and policy invariance are not equivalent: a representation may match across views while the action decoder still changes, or it may preserve view-specific information that is useful but should not alter the action for an equivalent state. We therefore place consistency at the action-flow output of the policy rather than at a fixed hidden representation; feature-level alternatives we explored (Appendix G) did not yield a reliable camera-track gain.

3 Cross-View Action-Flow Consistency

Action-equivalent pairs. Let s be the physical task state, c a scene-camera configuration, and $o = h(s, c)$ the rendered RGB observation. A training pair is

$$(o_0, o_p, l, q, a), \quad o_0 = h(s, c_0), \quad o_p = h(s, c_p), \quad (1)$$

where c_0 is the nominal camera, c_p is a perturbed camera, l is the language instruction, q is the proprioceptive state, and $a \in \mathbb{R}^{H \times A_{\text{act}}}$ is the demonstrated action chunk, where H is the prediction horizon and A_{act} is the number of active action dimensions. The two images differ only by scene camera; the task state, instruction, proprioception, and action target are shared.

Flow-level consistency. We use the convention that $t = 1$ is noise and $t = 0$ is the demonstrated action. x_t denotes the action sample at flow time t , and u_t the corresponding conditional velocity target. With $\varepsilon \in \mathbb{R}^{H \times A_{\text{act}}}$ drawn from $\mathcal{N}(0, I)$,

$$x_t = t\varepsilon + (1 - t)a, \quad u_t = \varepsilon - a. \quad (2)$$

For each action-equivalent pair, both views share a , x_t , and u_t . For K sampled flow-time/noise points $\{(t_k, \varepsilon_k)\}_{k=1}^K$, define

$$\begin{aligned} x_{t_k} &= t_k \varepsilon_k + (1 - t_k)a, & u_{t_k} &= \varepsilon_k - a, \\ v_0^k &= v_\theta(x_{t_k}, t_k \mid o_0, l, q), & v_p^k &= v_\theta(x_{t_k}, t_k \mid o_p, l, q). \end{aligned} \quad (3)$$

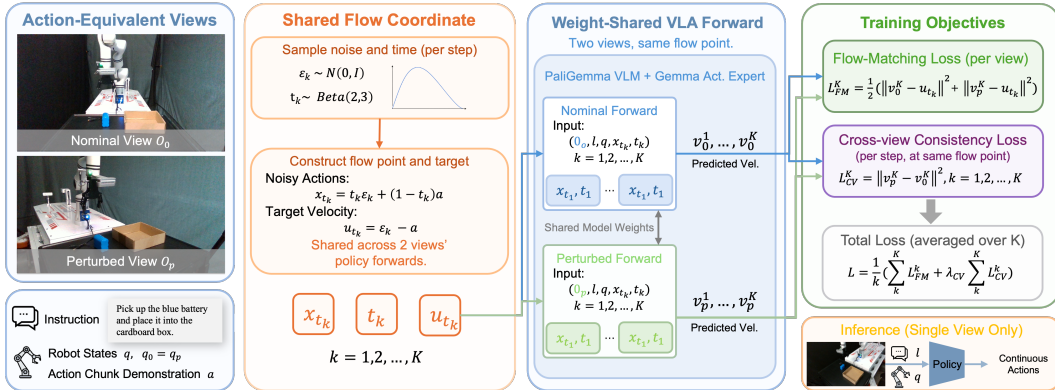


Figure 2: Cross-view action-flow consistency. During training, an action-equivalent nominal/perturbed view pair is processed by weight-shared VLA forward passes. Both views remain supervised by the standard flow-matching objective, while a cross-view loss encourages their action-flow predictions to agree at the same sampled flow coordinates. The pair is used only during training; inference uses a single RGB scene image, language instruction, and proprioceptive state.

Both views are anchored to the demonstrated action through standard flow matching:

$$\mathcal{L}_{\text{FM}}^k = \frac{1}{2} (\|v_0^k - u_{t_k}\|^2 + \|v_p^k - u_{t_k}\|^2). \quad (4)$$

The cross-view term compares the two velocity predictions on active action dimensions:

$$\mathcal{L}_{\text{CV}}^k = \|v_p^k[\cdot, : A_{\text{act}}] - v_0^k[\cdot, : A_{\text{act}}]\|^2. \quad (5)$$

The per-pair training objective is

$$\mathcal{L} = \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{FM}}^k + \lambda_{\text{CV}} \frac{1}{K} \sum_{k=1}^K \mathcal{L}_{\text{CV}}^k. \quad (6)$$

where $\lambda_{\text{CV}} > 0$ is a scalar weight controlling the strength of the cross-view regularization. The same (t_k, ε_k) is shared across both views within each pair, and gradients flow through both branches bilaterally.

Mean-residual reformulation. Writing the two-view predictions in mean-residual coordinates $\bar{v}_k = \frac{1}{2}(v_0^k + v_p^k)$ and $\delta_k = \frac{1}{2}(v_0^k - v_p^k)$, the per-pair objective in (6) admits the equivalent form

$$\mathcal{L}^k = \|\bar{v}_k - u_{t_k}\|^2 + (1 + 4\lambda_{\text{CV}}) \|\delta_k\|^2, \quad (7)$$

using $\frac{1}{2}(\|v_0^k - u_{t_k}\|^2 + \|v_p^k - u_{t_k}\|^2) = \|\bar{v}_k - u_{t_k}\|^2 + \|\delta_k\|^2$ and $\|v_p^k - v_0^k\|^2 = 4\|\delta_k\|^2$. The demonstrated action enters only through the mean prediction $\bar{v}_k$, while the cross-view term acts purely on the view-disagreement residual δ_k . Adding $\mathcal{L}_{\text{CV}}$ therefore does not redirect either view away from the demonstrated action; it increases the penalty on view-specific velocity residuals beyond standard flow matching, by an additional factor of $4\lambda_{\text{CV}}$. Two consequences follow: $\mathcal{L}_{\text{CV}}$ is not redundant with flow matching even when both views are anchored to the same target, and its effect concentrates on the part of the velocity field that is sensitive to scene-camera pose. For notational clarity, this decomposition is stated on the active action dimensions $[\cdot, : A_{\text{act}}]$; padded dimensions, when present, receive only the ordinary flow-matching term.

$\mathcal{L}_{\text{CV}}$ is a behavior-level consistency signal. Feature-level consistency does not guarantee action agreement (Appendix G), and final-action matching would require backpropagation through the numerical sampler, entangling local velocity disagreement with integration effects. We therefore regularize the velocity field itself, the quantity directly integrated to produce action chunks. Under the usual local Lipschitz assumption, velocity disagreement along the integration trajectory bounds the divergence of the integrated action chunks. Letting $a_0^*(s)$ and $a_p^*(s)$ denote the action chunks obtained by Euler-integrating v_θ from the nominal and perturbed views respectively, and $x_t^{(0)}$ a point on the nominal integration trajectory,

$$\mathbb{E}_{s,\varepsilon} [\|a_0^*(s) - a_p^*(s)\|^2] \leq C \mathbb{E}_{s,t,\varepsilon} [\|v_\theta(x_t^{(0)}, t | o_0, l, q) - v_\theta(x_t^{(0)}, t | o_p, l, q)\|^2], \quad (8)$$

where C depends on the sampler grid and the Lipschitz constant (Appendix E). The training-time loss $\mathcal{L}_{\text{CV}}$ in (5) evaluates this disagreement at the conditional flow-matching interpolation points $x_{t_k} = t_k \varepsilon_k + (1 - t_k) a$, which coincide with the integration trajectory under the optimal flow-matching solution and otherwise serve as a tractable surrogate for the right-hand side. The loss therefore targets the velocity field whose integration yields rollout behavior, not an auxiliary representation.

Shuffled pairs are the mechanism control. A key control keeps flow matching intact while breaking action equivalence only inside $\mathcal{L}_{\text{CV}}$. For a batch of N pairs indexed by i , let π be a derangement of $\{1, \dots, N\}$. The shuffled cross-view term replaces the matched perturbed prediction with one from a different state:

$$\mathcal{L}_{\text{CV,shuf}}^k = \|v_{p,\pi(i)}^k[\cdot, : A_{\text{act}}] - v_{0,i}^k[\cdot, : A_{\text{act}}]\|^2. \quad (9)$$

Every view still receives its correct row-local flow-matching target. At the population level, suppressing shared arguments for readability,

$$\mathbb{E}_{s,s'} \|v_p(s') - v_0(s)\|^2 = \mathbb{E}_s \|v_p(s) - v_0(s)\|^2 + 2 \text{Tr}(\text{Cov}_s(v_p(s), v_0(s))). \quad (10)$$

The second term is state-marginal covariance, not view disagreement for the same physical state (proof in Appendix E). When v_p and v_0 retain meaningful state-dependent action information, this covariance is typically positive for meaningful state-dependent policies; minimizing the shuffled objective therefore actively opposes the supervised flow-matching signal by pulling per-state velocity predictions toward a common state-marginal mean. A shuffled-pair control thus tests whether the cross-view loss depends on true action-equivalent pairing or acts merely as a generic smoothing penalty.

4 Experiments

We evaluate whether cross-view action-flow consistency improves scene-camera viewpoint robustness in simulation and on a real robot, using the LIBERO-Plus camera-perturbation benchmark and a held-out camera placement protocol respectively. The simulation experiments use same-state rerendered LIBERO pairs for training and the official LIBERO-Plus camera-perturbation benchmark for evaluation. The real-robot experiment uses synchronized multi-camera demonstrations for training and evaluates deployment from a held-out camera placement not used during data collection. In all experiments, the wrist-camera stream is masked during both training and evaluation: the wrist view is not subject to the scene-camera perturbation, so allowing the policy to use it would introduce an unperturbed visual shortcut that confounds the causal interpretation of scene-camera robustness. This isolates scene-camera attribution but makes our absolute success rates incomparable to standard LIBERO-Plus numbers that retain the wrist view; all comparisons reported here are made within this single-scene-RGB protocol against identically evaluated baselines.

4.1 Experimental setup

Simulation training data. We construct action-equivalent pairs from the original LIBERO demonstrations [15] by resetting the simulator to stored MuJoCo states and rerendering the same physical state under a nominal scene camera and a perturbed scene camera (Figure 3a). Each pair shares the task, language instruction, robot state, and demonstrated actions; only the scene-camera pose changes. Perturbed cameras are sampled from the LIBERO-Plus C1/C2/C3 distribution [16] (distance and scale, spherical position, endpoint orientation); training and evaluation share this distribution family but use disjoint samples, as the evaluation rollouts in Section 4.2 come from the separate LIBERO-Plus benchmark split and no evaluation camera poses, task initial states, or action labels enter training. We generate 338,575 same-state pairs from `libero_spatial`, `libero_object`, `libero_goal`, and `libero_100`, and use the resulting pair data for the final simulation policies.

Implementation details. All policies are fine-tuned from the $\pi_{0.5}$ checkpoint for 10,000 steps. The proposed method uses $K = 2$ flow samples per pair, $\lambda_{CV} = 0.10$, and $t_k \sim \text{Beta}(2, 3)$; the rationale for this flow-time concentration is given in Appendix F.

Evaluation protocol. The LIBERO-Plus camera-perturbation benchmark [16] contains 1,599 camera-perturbation task instances. It is divided into C1 camera distance and scale ($n = 313$), C2 spherical camera position ($n = 992$), and C3 endpoint camera orientation ($n = 294$). We run 4,797 trials, corresponding to 3 full rollouts for each individual evaluation. We also report performance on the standard LIBERO benchmark (denoted as ID) with nominal cameras (4 suites $\times$ 30 trials); all policies use the single-scene-RGB, wrist-masked condition described above.

Comparisons. The nominal-only baseline fine-tunes the open-sourced base $\pi_{0.5}$ policy on standard LIBERO nominal-camera data and measures fixed-camera brittleness. The naive mixed-camera SFT baseline fine-tunes on LIBERO-Plus public camera-perturbation data with ordinary flow matching. The primary same-data control is FM-only training on our same-state pair dataset, which uses the same rendered frames, architecture, optimizer, and 10,000-step budget as our method but sets $\lambda_{CV} = 0$. Ablations then modify the cross-view term, augmentation geometry, gradient coupling, or pair identity.

181 4.2 Main simulation results on LIBERO-Plus

Table 1: Main simulation results on LIBERO-Plus. ID: standard LIBERO nominal-camera benchmark. Camera: LIBERO-Plus camera-perturbation aggregate. Both the FM-only same-state pair-data control and the proposed method are reported as mean $\pm$ std over training seeds 42–44. The FM-only pair-data row differs from the proposed method only in $\lambda_{CV} = 0$.

Method	Training data / objective	ID	Camera	C1	C2	C3
Nominal-only baseline	nominal LIBERO, FM only	85.8	16.8	1.1	13.2	45.7
Naive mixed-camera SFT	LIBERO-Plus camera data, FM only	69.2	74.7	68.4	75.7	78.2
FM-only on same-state pairs, seeds 42–44	rerendered pairs, FM only	95.0 $\pm$ 4.3	79.8 $\pm$ 0.8	72.6 $\pm$ 0.2	79.9 $\pm$ 1.5	87.2 $\pm$ 0.7
Proposed, seeds 42–44	rerendered pairs, bilateral $K = 2$	95.0 $\pm$ 0.8	87.2$\pm$0.4	81.8$\pm$1.0	87.9$\pm$0.5	90.6$\pm$0.9

182 Table 1 summarizes the main results; the full aggregate table and per-seed results are in Appendix B
 183 and Appendix C, respectively. The proposed method reaches 87.2 $\pm$ 0.4% on the camera track, with
 184 +7.4pp over the FM-only same-data control (79.8 $\pm$ 0.8%) and +12.5pp over naive mixed-camera
 185 SFT. Both pair-data variants are trained from identical data and differ only in λ_{CV} ; the gap is consis-
 186 tent across seeds (Appendix C). Both methods preserve nominal-camera ID performance at 95.0%
 187 mean, with the proposed method showing substantially lower per-seed variance ($\pm$ 0.8 vs. $\pm$ 4.3).
 188 The camera-track gains are largest on C1 (+9.2pp) and C2 (+8.0pp), and remain positive on C3
 189 (+3.4pp), where the proposed method reaches 90.6 $\pm$ 0.9% compared with 87.2 $\pm$ 0.7% for FM-only.

190 4.3 Mechanism analysis and ablations

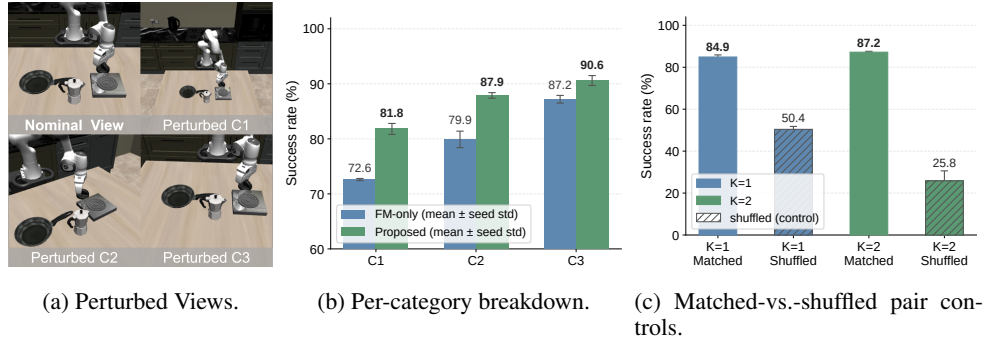


Figure 3: LIBERO-Plus camera-track analysis. (a) Rendered nominal and perturbed scene-camera views. (b) Per-category success rates of FM-only vs. the proposed method, both shown as mean $\pm$ seed std over seeds 42–44. (c) Matched vs. shuffled pair controls for $K = 1$ and $K = 2$; shuffled bars break action-equivalent pairing inside $\mathcal{L}_{CV}$ while keeping per-view flow-matching targets correct.

191 Table 2 isolates the source of the camera-track improvement, with supporting evidence in Figure 3.
 192 Removing the cross-view term while preserving the same pair data drops camera-track performance
 193 from 87.2 $\pm$ 0.4% to 79.8 $\pm$ 0.8%, a 7.4pp gap attributable solely to λ_{CV} (per-seed results in Ap-
 194 pendix C). Because both variants share the same data, this gap reflects $\mathcal{L}_{CV}$ alone, independent of
 195 any train/eval camera overlap.

196 A stronger test breaks action equivalence inside $\mathcal{L}_{CV}$ by shuffling pairs across the batch, so that the
 197 loss compares velocity predictions from different physical states. Per-view flow-matching targets
 198 remain correct, but the cross-view signal becomes incoherent. Performance collapses from 84.9%
 199 to 50.4% at $K = 1$, and from 87.2% to 25.8% at $K = 2$ (Figure 3c). Bilateral coupling and multi-
 200 sample averaging amplify matched and mismatched targets alike, so the recipe with the strongest
 201 matched result also collapses most severely under wrong pairing. This rules out generic smoothing
 202 of the velocity field as an explanation; our method specifically requires same-state correspondence
 203 to be useful.

204 A stop-gradient teacher–student variant achieves the highest in-distribution success (96.7%) but only
 205 84.3% on the camera track, 2.9pp below the bilateral recipe. This suggests that anchoring only one

Table 2: Ablations and controls isolating the source of the camera-track improvement. The stop-gradient row changes both gradient structure and flow-time distribution simultaneously; it therefore does not isolate either factor.

Question	Configuration	ID	Camera	Takeaway
Data exposure only?	FM-only on same-state pairs (3 seeds)	95.0 ± 4.3	79.8 ± 0.8	Camera-diverse pair data alone is insufficient.
One-sided coupling?	Stop-gradient $K=2$, Beta(1.5, 1)	96.7	84.3	High ID, but lower robustness than the bilateral recipe.
Wrong pairs?	Shuffled single-sample CV	80.8	50.4	Matching must be action-equivalent.
Wrong pairs?	Shuffled bilateral $K=2 + \text{Beta}(2, 3)$	50.8	25.8	Stronger wrong cross-view targets severely damage behavior.
Complete recipe	Proposed bilateral $K=2 + \text{Beta}(2, 3)$, seeds 42–44	95.0 ± 0.8	87.2 ± 0.4	Best observed robustness/ID tradeoff.

branch is a weaker form of cross-view coupling in this setting. Sensitivity to λ_{CV} is reported in Appendix B.

4.4 Real-robot held-out camera evaluation

We collect synchronized multi-camera demonstrations for three tabletop tasks: Pick & Place (blue battery into cardboard box), Contact Rich (closing a laptop lid), and Dexterous Operation (taking headphones off a stand). The three synchronized cameras at each timestep provide the real-hardware analog of the simulator-state-reset pairs in Section 4.2; independently collected trajectories from different placements are not used as pairs, since they are not action-equivalent.

At deployment, each policy receives one RGB scene image, language, and proprioception, with the wrist stream masked. We evaluate from three seen training-camera placements (C_0, C_1, C_2 in Figure 4) and three held-out placements never used during data collection: H_0 (a farther distance extrapolation from C_0), H_1 (an azimuth rotation around the table center between C_0 and C_1), and H_2 (a higher and more top-down placement between C_0 and C_2); the six placements are visualized in Figure 6 (Appendix D.3). We run 10 rollouts per task–placement cell (90 rollouts per method per condition).

Figure 5 reports aggregate and task-level results; full counts and Wilson intervals are in Appendix D. On held-out cameras, FM-only multi-view training reaches 48/90 aggregate successes, while the proposed objective reaches 67/90 (descriptive two-proportion test, $p < 0.005$; we treat per-task/per-placement counts as primary). The seen-camera condition shows that both methods can use views represented in the demonstration data (80/90 for FM-only, 79/90 for the proposed method); the held-out-camera gap separates the two training signals. Gains appear on all three tasks but their magnitudes are uneven: the Laptop lid task transfers nearly losslessly under the proposed objective (30/30 held-out), the Battery box task improves from 17/30 to 23/30, and the Headphone stand task carries the largest absolute gap, 5/30 to 14/30.

Two patterns emerge from the per-placement breakdown (Appendix D). First, the held-out gain grows with the geometric aggressiveness of the placement: the largest task–placement margins oc-

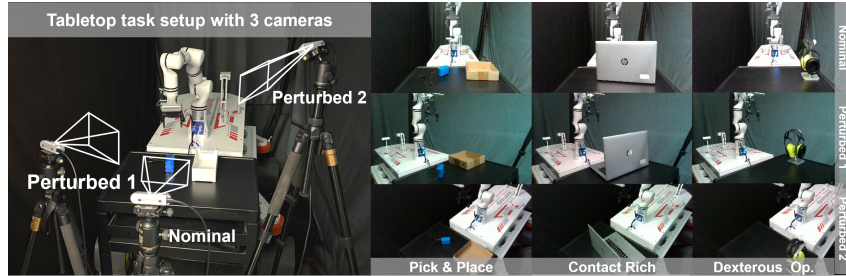


Figure 4: Real-robot synchronized multi-camera setup. Training demonstrations are collected with synchronized scene cameras C_0, C_1, C_2 to form action-equivalent pairs. At deployment, the policy receives only a single RGB scene image. The key evaluation condition places the deployment camera at a position never observed during training.

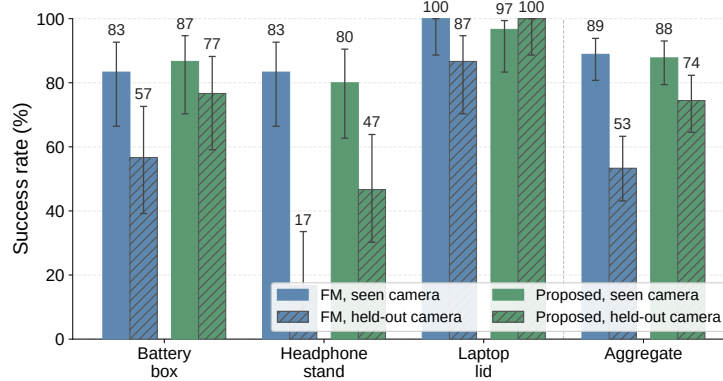


Figure 5: Task-level and aggregate success rates for seen and held-out camera placements.

cur on H_2 , the highest and most top-down held-out view. Second, the Headphone stand task on H_2 is the single most informative case: FM-only succeeds in 0/10 rollouts while the proposed objective succeeds in 4/10, suggesting that the cross-view objective adds the most value where held-out viewpoint is most distant from training. Even with this gain, 16/30 Headphone stand rollouts still fail: failures cluster around fine gripper-headphone alignment near contact, where held-out views occlude the thin band and small approach-angle errors cause mis-grasps (Appendix D).

5 Limitations

Paired training data. The method requires true action-equivalent pairs: in simulation by resetting MuJoCo states and re-rendering, on hardware by synchronized cameras observing the same physical state at the same timestamp. This requirement excludes most existing single-camera robot datasets, in which different views typically come from different rollouts. Approximate pairs constructed from independently collected trajectories are not a safe shortcut: by the shuffled-pair decomposition in Section 3, enforcing consistency between non-equivalent states pushes per-state velocity predictions toward a state-marginal mean and actively opposes the supervised flow-matching signal, as confirmed empirically by the collapse to 25.8% in Figure 3c. Reducing this is our direction for future work (Appendix I).

Observability under single RGB. Cross-view consistency improves action agreement across views of the same state but cannot recover information not visually present in the input RGB; when held-out cameras occlude task-critical geometry or compress depth cues near contact, the objective has nothing to anchor against. This matches our inference-contract choice (Appendix H), no depth, point clouds, RGB-D, or tactile sensing at deployment, and combining cross-view consistency with these is a direct extension.

Scope of evaluation. The hardware study is intentionally controlled: three tabletop tasks, static external scene cameras, three seen and three held-out placements, and 10 rollouts per task—placement cell. It does not establish robustness beyond LIBERO-Plus C1/C2/C3 shifts, moving cameras, lighting changes, clutter, transparent/deformable objects, or mobile-base manipulation. Extending the evaluation protocol to these conditions, as well as transferring the action-flow consistency objective to discrete-token VLA backbones and other action-head families, is left to future work.

6 Conclusion

Camera motion can change a VLA’s action predictions even when the physical task state is unchanged. For flow-based VLAs, we address this by regularizing the action-flow velocity field across action-equivalent views; on LIBERO-Plus and real-robot held-out-camera evaluations, this training-only objective improves over nominal-only, camera-diverse SFT, and same-data FM-only baselines while preserving single-scene-RGB inference.

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A Dataset and Evaluation Details

A.1 Same-state simulation pair construction

We use original LIBERO demonstrations as the source for same-state pair construction. For a selected demonstration timestep t , we load the corresponding environment, reset the simulator to the stored MuJoCo state, call `sim.forward()`, and render two scene-camera observations: a nominal view and a perturbed view. The row metadata records suite, task, episode, timestep, state hash, action-chunk hash, camera category, and camera parameters. A matched pair is valid only if the two slots share the same simulator state, action chunk, language instruction, and proprioceptive state.

The generated dataset contains 338,575 same-state pairs from `libero_spatial`, `libero_object`, `libero_goal`, and `libero_100`, where `libero_100` consists of `libero_10` and `libero_90`. Camera categories follow LIBERO-Plus: C1 distance/scale, C2 spherical camera position, and C3 end-point orientation. The real-robot dataset follows the same logical structure, but exact same-state observations are obtained from synchronized cameras rather than simulator reset.

A.2 Train/evaluation separation

The simulation training pairs are generated from original LIBERO demonstration states, with perturbed scene cameras drawn in line with the camera distribution of LIBERO-Plus’s official training data under the C1/C2/C3 taxonomy. Evaluation is performed on the official LIBERO-Plus camera-perturbation benchmark rollouts, which constitute a separate split: no evaluation camera poses, task initial states, or action labels enter training, and we do not evaluate on a held-out split of the rendered training-pair dataset itself. The reported simulation results therefore measure generalization across LIBERO-Plus’s own train/evaluation split rather than performance on training-time camera instances. No wrist image is used in any reported simulation policy, because LIBERO-Plus perturbs the scene camera but leaves wrist observations stable; retaining wrist images would no longer isolate scene-camera robustness. All training and evaluation use a $6\times$ NVIDIA RTX PRO 6000 Server Edition GPU setup; detailed hyperparameters are documented in the open-sourced code repository linked from the project page.

A.3 Rollout metrics

All rollout success rates are binomial proportions. Single-seed baseline and ablation rows are reported with rollout-level Wilson 95% confidence intervals. The primary FM-only same-state-pair control and the proposed method are reported as mean $\pm$ standard deviation over training seeds 42–44; their seed-level results are given in Appendix C.

B Full Simulation Tables

Table 3: Full simulation table. Single-seed baseline rows are reported with rollout-level Wilson 95% confidence intervals. The FM-only same-state pair control and the proposed method are both reported as mean $\pm$ standard deviation across training seeds 42–44; per-seed values for both are in Table 5.

Method	ID	Camera	C1	C2	C3
Nominal-only baseline	85.8 [78.5, 91.0]	16.8 [15.8, 17.9]	1.1 [0.6, 1.9]	13.2 [12.0, 14.5]	45.7 [42.4, 49.0]
Naive mixed-camera SFT	69.2 [60.4, 76.7]	74.7 [73.4, 75.9]	68.4 [65.3, 71.3]	75.7 [74.1, 77.2]	78.2 [75.4, 80.8]
FM-only on same-state pairs, seeds 42–44	95.0 $\pm$ 4.3	79.8 $\pm$ 0.8	72.6 $\pm$ 0.2	79.9 $\pm$ 1.5	87.2 $\pm$ 0.7
Proposed, seeds 42–44	95.0 $\pm$ 0.8	87.2 $\pm$ 0.4	81.8 $\pm$ 1.0	87.9 $\pm$ 0.5	90.6 $\pm$ 0.9

On the camera track, $\lambda_{CV}=0.05$ and 0.10 lie within overlapping rollout-level Wilson intervals (81.1 [80.0, 82.2] vs. 80.9 [79.8, 82.0]); camera-track performance is thus insensitive to λ_{CV} across $[0.05, 0.10]$ in this regime, and we carry $\lambda_{CV}=0.10$ to the final configuration.

Table 4: Extended ablations. The λ_{CV} sweep is run in an exploratory single-sample, independent-augmentation regime, distinct from the final $K=2$ pair-consistent configuration in which $\lambda_{CV}=0.10$ was selected; its absolute numbers are therefore not directly comparable to Table 1.

Configuration	ID	Camera
<i>λ_{CV} sensitivity, single-sample symmetric, independent spatial augmentation</i>		
$\lambda = 0.05$	96.7 [91.7, 98.7]	81.1 [80.0, 82.2]
$\lambda = 0.10$	90.0 [83.3, 94.2]	80.9 [79.8, 82.0]
$\lambda = 0.20$	85.8 [78.5, 91.0]	73.6 [72.5, 74.7]
$\lambda = 0.50$	82.5 [74.7, 88.3]	68.1 [67.0, 69.2]
<i>Pair augmentation regime, single-sample symmetric, $\lambda = 0.10$</i>		
Independent spatial augmentation	90.0 [83.3, 94.2]	80.9 [79.8, 82.0]
Pair-consistent spatial handling	92.5 [86.4, 96.0]	84.9 [83.9, 85.9]
<i>Matched vs. shuffled, pair-consistent</i>		
Shuffled single-sample symmetric	80.8 [72.9, 86.9]	50.4 [49.0, 51.8]
Shuffled bilateral $K = 2 + \text{Beta}(2, 3)$	50.8 [42.0, 59.6]	25.8 [24.4, 27.2]

C Seed-Level Results for FM-only and Proposed

We train both the FM-only same-state pair control ($\lambda_{CV} = 0$) and the proposed bilateral $K = 2 + \text{Beta}(2, 3)$ configuration with three independent training seeds (42, 43, 44) and evaluate each on the full LIBERO-Plus camera-perturbation track and the LIBERO ID benchmark. The two methods share identical training data, architecture, optimizer, and step budget; they differ only in λ_{CV} .

Table 5: Per-seed results for the FM-only same-state pair control and the proposed bilateral $K = 2 + \text{Beta}(2, 3)$ configuration. Camera denotes the LIBERO-Plus camera-perturbation aggregate.

Method	Seed	ID	Camera	C1	C2	C3
FM-only on same-state pairs	42	92.5	79.5	72.8	79.4	86.6
	43	100.0	80.7	72.5	81.5	87.0
	44	92.5	79.2	72.6	78.7	88.0
	Mean $\pm$ std.	95.0 $\pm$ 4.3	79.8 $\pm$ 0.8	72.6 $\pm$ 0.2	79.9 $\pm$ 1.5	87.2 $\pm$ 0.7
Proposed, bilateral $K = 2 + \text{Beta}(2, 3)$	42	95.0	86.9	80.7	88.1	89.7
	43	94.2	87.6	82.2	88.3	90.7
	44	95.8	87.2	82.6	87.3	91.5
	Mean $\pm$ std.	95.0 $\pm$ 0.8	87.2 $\pm$ 0.4	81.8 $\pm$ 1.0	87.9 $\pm$ 0.5	90.6 $\pm$ 0.9

Across seeds, the proposed method’s per-seed Camera score (lowest 86.9%, highest 87.6%) is strictly above the FM-only same-state pair control’s per-seed Camera score (lowest 79.2%, highest 80.7%). The matched-seed gap ranges from +6.9pp (seed 43) to +8.0pp (seed 44); moreover, the lowest proposed seed still exceeds the highest FM-only seed by +6.2pp. Category-level gains are positive on all three camera perturbation types: +9.2pp on C1, +8.0pp on C2, and +3.4pp on C3. The two methods reach the same ID mean (95.0%), but FM-only exhibits higher per-seed variance (± 4.3 vs. ± 0.8): the FM-only ID ranges from 92.5% to 100.0%, while the proposed method stays within a [94.2, 95.8] band. We treat this ID-variance reduction as a secondary observation rather than a primary claim, since ID uses fewer rollouts than the LIBERO-Plus camera track.

D Real-Robot Experiments and Full Tables

D.1 Hardware setup and synchronization

Real-robot demonstrations are collected with three synchronized external scene cameras C_0, C_1, C_2 . For a demonstration timestep t , the tuple $(C_0[t], C_1[t], C_2[t], l, q[t], a[t])$ is treated as action-equivalent supervision because all cameras observe the same physical state at the same timestamp. Trajectories collected independently from different camera placements are not used for the cross-view term, because they do not guarantee same-state action equivalence. Table 6 summarizes the real-robot data collection, training, and checkpoint-selection protocol. Held-out camera placements are reserved strictly for evaluation and are not used for training or checkpoint selection.

Table 6: Real-robot data collection and training details. Held-out camera placements are used only for evaluation and never for training or checkpoint selection.

Item	Setting
Robot and action space	RealMan RM-75 7-DoF manipulator; delta end-effector pose actions, controlled at 15 Hz.
Tasks	Battery box; Headphone stand; Laptop lid.
Training cameras	Three synchronized external scene cameras, denoted C_0 , C_1 , and C_2 .
Held-out cameras	H_0 , H_1 , and H_2 ; used only for evaluation.
Demonstrations	70 demonstrations per task, collected with domain randomization.
Training samples	Each demonstration timestep is recorded by three synchronized cameras. Action-equivalent training pairs are constructed from these synchronized views.
Image preprocessing	Intel RealSense D435i cameras; RGB images are captured at 640×480 resolution and 15 fps.
Initialization	$\pi_{0.5}$ checkpoint.
FM-only training	Same real-robot multi-view data and training budget as the proposed method, with $\lambda_{CV} = 0$. Training steps, batch size, optimizer, and learning-rate schedule follow the simulation setting.
Proposed training	Cross-view action-flow consistency on synchronized action-equivalent pairs. Hyperparameters follow the simulation setting.
Checkpoint selection	Final checkpoint after training; no held-out-camera rollouts are used for checkpoint selection.
Evaluation budget	10 rollouts per task–placement cell; 90 seen-camera and 90 held-out-camera rollouts per method.

425 D.2 Tasks and rollout budget

426 The three tasks are: blue battery into cardboard box, take the headphones off the stand, and close
 427 laptop lid. For each method and deployment camera condition, we run 30 rollouts per task (3 camera
 428 placements $\times$ 10 rollouts each). With two core methods and two deployment camera conditions, the
 429 total core evaluation is $2 \times 3 \times 2 \times 30 = 360$ rollouts.

Table 7: Real-robot task-level breakdown. Each cell reports success rate with Wilson 95% confidence interval and success count.

Method	Camera	Battery box	Headphone stand	Laptop lid	Aggregate
FM, multi-view mixed	Seen	83.3 [66.4, 92.7] (25/30)	83.3 [66.4, 92.7] (25/30)	100 [88.7, 100] (30/30)	88.9 [80.7, 93.8] (80/90)
FM, multi-view mixed	Held-out	56.7 [38.9, 72.9] (17/30)	16.7 [7.3, 33.6] (5/30)	86.7 [70.3, 94.7] (26/30)	53.3 [43.1, 63.3] (48/90)
Proposed	Seen	86.7 [70.3, 94.7] (26/30)	80.0 [62.7, 90.5] (24/30)	96.7 [83.3, 99.4] (29/30)	87.8 [79.4, 93.0] (79/90)
Proposed	Held-out	76.7 [59.1, 88.2] (23/30)	46.7 [30.2, 63.9] (14/30)	100 [88.7, 100] (30/30)	74.4 [64.6, 82.4] (67/90)

430 D.3 Per-placement task-level results

431 Table 8 expands the aggregate results in Table 7 to per-placement task-level success counts. H_0
 432 extrapolates camera distance away from C_0 ; H_1 rotates azimuthally around the table center between
 433 C_0 and C_1 ; H_2 raises and tilts the view between C_0 and C_2 .

434 Two patterns emerge, both visually consistent with Figure 6. First, the proposed method’s advantage
 435 scales with the geometric aggressiveness of the held-out placement: aggregating across tasks, FM-
 436 only and the proposed method differ by 4 rollouts on H_0 (20/30 vs 24/30), 6 on H_1 (17/30 vs
 437 23/30), and 9 on H_2 (11/30 vs 20/30). Second, the Headphone stand task on H_2 is the most extreme
 438 case: FM-only never succeeds (0/10), while the proposed method succeeds in 4/10 rollouts—
 439 consistent with the structural-observability limitation discussed in Section 5, since H_2 combines the
 440 most aggressive viewpoint change with the task that has the smallest visually-recoverable contact
 441 geometry.

Table 8: Per-placement task-level success counts (out of 10 rollouts per cell). Seen placements are C_0, C_1, C_2 ; held-out placements are H_0, H_1, H_2 .

Task	Method	Seen			Held-out		
		C_0	C_1	C_2	H_0	H_1	H_2
Battery box	FM	9	8	8	8	6	3
	Proposed	10	9	7	9	8	6
Laptop lid	FM	10	10	10	9	9	8
	Proposed	10	10	9	10	10	10
Headphone stand	FM	8	9	8	3	2	0
	Proposed	7	9	8	5	5	4

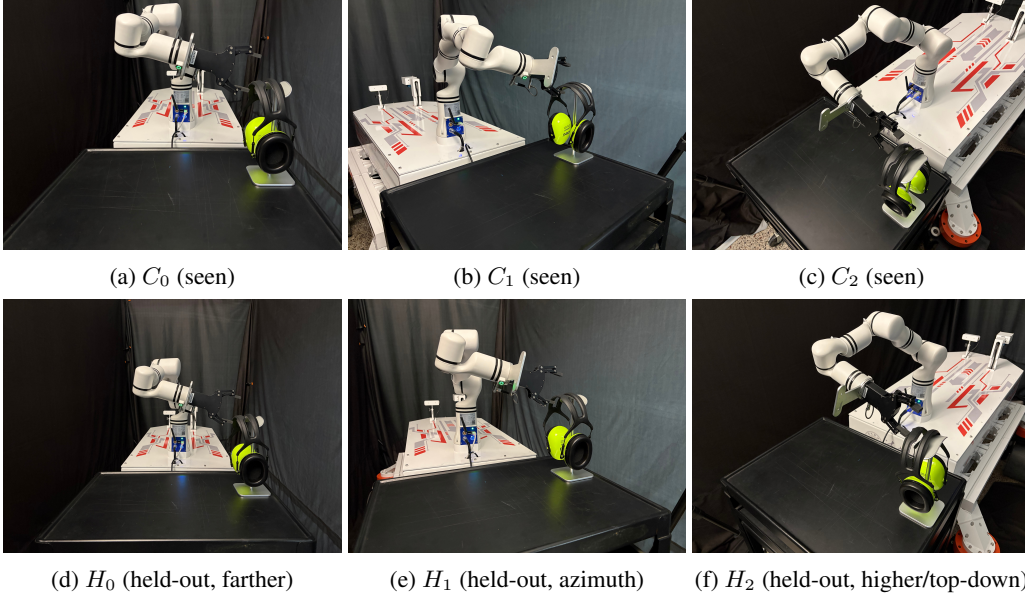


Figure 6: Scene-camera views used for the real-robot held-out evaluation. All six images show the same physical state of the Headphone stand task moments before grasp contact, with the gripper positioned a few centimeters above the headphone band. Top row: training-camera placements C_0, C_1, C_2 used for both demonstration collection and cross-view supervision. Bottom row: held-out placements H_0, H_1, H_2 used only at evaluation. The held-out views progressively depart from the training cameras: H_0 extrapolates camera distance, H_1 rotates azimuthally, and H_2 combines a higher position with a more top-down angle.

442 D.4 Real-robot reporting rule

443 The main real-robot claim should be based on held-out-camera success, the seen-to-held-out drop,
 444 and the number of tasks that improve. We report real-robot results as success counts and Wilson
 445 intervals. Qualitative rollout frames can be shown in the real-robot figure, videos are supplementary
 446 evidence and visible at project page.

447 D.5 Qualitative failure analysis

448 The held-out-camera degradation is task-dependent. The headphone-stand task shows the largest
 449 drop for both methods: FM-only multi-view mixed training reaches only 5/30 held-out successes,
 450 while the proposed method improves to 14/30 but remains far from saturated. Qualitatively, this
 451 task requires fine gripper-headphone alignment and is sensitive to depth ambiguity under novel
 452 viewpoints. The laptop-lid task is visually larger and more constrained by contact geometry, and
 453 both methods transfer more reliably, with the proposed method reaching 30/30 held-out successes.
 454 These patterns suggest that cross-view action-flow consistency helps substantially under camera
 455 shift, but does not remove failures caused by poor observability, thin objects, or viewpoint-induced
 456 depth ambiguity.

457 E Mechanism Proof

458 This section expands the mechanism statements used in the main paper. It is not required to under-
459 stand the main experimental claim; the main text contains the key equations and empirical controls.

460 **Proposition 1** (Action-chunk divergence bound). *Let $0 = t_0 < t_1 < \dots < t_N = 1$ be a uniform
461 Euler grid with step size $\Delta t = 1/N$. Let $x_{t_n}^{(0)}$ and $x_{t_n}^{(p)}$ denote the Euler-integration trajectories for
462 the nominal and perturbed views respectively, both initialized at the same noise $x_{t_N}^{(0)} = x_{t_N}^{(p)} = \varepsilon$. If
463 $v_\theta(\cdot, t \mid o, l, q)$ is L -Lipschitz in its first argument uniformly in t , then*

$$\|x_{t_0}^{(0)} - x_{t_0}^{(p)}\|^2 \leq e^{2L} \sum_{n=1}^N \Delta t \|v_\theta(x_{t_n}^{(0)}, t_n \mid o_0, l, q) - v_\theta(x_{t_n}^{(0)}, t_n \mid o_p, l, q)\|^2. \quad (11)$$

464 *Proof.* Define $\delta_n = \|x_{t_n}^{(0)} - x_{t_n}^{(p)}\|$ and

$$e_n = \|v_\theta(x_{t_n}^{(0)}, t_n \mid o_0, l, q) - v_\theta(x_{t_n}^{(0)}, t_n \mid o_p, l, q)\|.$$

465 The Euler update from t_n to t_{n-1} is $x_{t_{n-1}}^{(\cdot)} = x_{t_n}^{(\cdot)} - \Delta t v_\theta(x_{t_n}^{(\cdot)}, t_n \mid o, l, q)$. Subtracting the two
466 trajectories and applying the triangle inequality:

$$\delta_{n-1} \leq \delta_n + \Delta t \|v_\theta(x_{t_n}^{(0)}, t_n \mid o_0) - v_\theta(x_{t_n}^{(p)}, t_n \mid o_p)\|. \quad (12)$$

467 Inserting $\pm v_\theta(x_{t_n}^{(0)}, t_n \mid o_p)$ into the norm and applying the L -Lipschitz condition to the second
468 difference:

$$\delta_{n-1} \leq (1 + L\Delta t) \delta_n + \Delta t e_n. \quad (13)$$

469 Unrolling (13) from $n = N$ down to $n = 1$ with $\delta_N = 0$ gives

$$\delta_0 \leq \sum_{n=1}^N (1 + L\Delta t)^{n-1} \Delta t e_n \leq e^L \sum_{n=1}^N \Delta t e_n, \quad (14)$$

470 where $(1 + L\Delta t)^{n-1} \leq (1 + L\Delta t)^N \leq e^{LN\Delta t} = e^L$. Squaring (14) and applying Cauchy–Schwarz
471 with uniform weights Δt :

$$\delta_0^2 \leq e^{2L} \left(\sum_{n=1}^N \Delta t e_n \right)^2 \leq e^{2L} \underbrace{\left(\sum_{n=1}^N \Delta t \right)}_{=1} \left(\sum_{n=1}^N \Delta t e_n^2 \right) = e^{2L} \sum_{n=1}^N \Delta t e_n^2, \quad (15)$$

472 which is (11). □

473 **Remark 1** (Training-time surrogate). *Proposition 1 bounds action-chunk divergence by velocity
474 disagreement evaluated at the nominal Euler trajectory points $x_{t_n}^{(0)}$. The cross-view loss $\mathcal{L}_{CV}$ in (5)
475 evaluates disagreement at the training-time linear interpolation points $x_{t_k} = t_k \varepsilon + (1 - t_k) a$, which
476 differ from $x_{t_n}^{(0)}$ in general. The two sets of points coincide when the learned velocity field pro-
477 duces straight-line trajectories, which holds for the optimal conditional flow matching solution. In
478 practice, the training-time objective therefore serves as a tractable surrogate for the right-hand side
479 of (11); this is the sense in which it is a “local surrogate” as described in the main text. The bound
480 is intended as a qualitative motivation rather than a tight quantitative estimate: the constant can be
481 loose for high-capacity neural velocity fields. Its role is to show that controlling velocity disagree-
482 ment along the integration path is sufficient to control action-chunk divergence, while the training
483 objective provides a practical surrogate for that quantity.*

484 **Lemma 1** (Shuffled-loss decomposition). *Let s and s' be independent draws from the same state
485 distribution. Omitting shared arguments (x_t, t, l, q) for readability, the population shuffled cross-
486 view loss satisfies*

$$\mathbb{E}_{s, s'} \|v_p(s') - v_0(s)\|^2 = \mathbb{E}_s \|v_p(s) - v_0(s)\|^2 + 2 \text{Tr}(\text{Cov}_s(v_p(s), v_0(s))). \quad (16)$$

487 *Proof.* Expanding the squared norm and using independence of s and s' :

$$\mathbb{E}_{s,s'} \|v_p(s') - v_0(s)\|^2 = \mathbb{E} \|v_p\|^2 + \mathbb{E} \|v_0\|^2 - 2 \langle \mathbb{E} v_p, \mathbb{E} v_0 \rangle, \quad (17)$$

488 where the cross term factorizes as $\mathbb{E}_{s,s'} [v_p(s')^\top v_0(s)] = (\mathbb{E}_{s'} v_p(s'))^\top (\mathbb{E}_s v_0(s))$ by independence.
 489 The matched loss expands as

$$\mathbb{E}_s \|v_p(s) - v_0(s)\|^2 = \mathbb{E} \|v_p\|^2 + \mathbb{E} \|v_0\|^2 - 2 \mathbb{E} \langle v_p, v_0 \rangle. \quad (18)$$

490 Subtracting (18) from (17):

$$\mathbb{E}_{s,s'} \|v_p(s') - v_0(s)\|^2 - \mathbb{E}_s \|v_p(s) - v_0(s)\|^2 = 2(\mathbb{E} \langle v_p, v_0 \rangle - \langle \mathbb{E} v_p, \mathbb{E} v_0 \rangle) = 2 \text{Tr}(\text{Cov}_s(v_p, v_0)), \quad (19)$$

491 which gives (16). $\square$

492 F Implementation and Reproducibility Details

493 **Backbone and action expert.** All policies are $\pi_{0.5}$ -style models with the standard pretrained
 494 vision-language trunk and a flow-matching action expert [5]. The simulation action space uses the
 495 LIBERO 7-DoF action representation. Although the underlying action representation may contain
 496 padded dimensions, the cross-view consistency term is applied only to the active action dimensions,
 497 $A_{\text{act}} = 7$.

498 **Training schedule.** Unless otherwise stated, simulation policies are trained for 10,000 steps from
 499 the same pretrained initialization, with the same optimizer, learning-rate schedule, batch size, and
 500 worker configuration. All training configs are provided in the open-sourced code repository linked
 501 from the project page. The primary FM-only same-state-pair control and the proposed configuration
 502 are both evaluated over seeds 42, 43, and 44. Final simulation evaluation uses the checkpoint at the
 503 end of training.

504 **Pair-consistent augmentation.** For paired cross-view training, random spatial transformations
 505 must not be sampled independently for the two paired slots. Independent crops or affine transforma-
 506 tions can introduce artificial pixel-space disagreement unrelated to camera pose and can consume
 507 the cross-view loss on augmentation artifacts. The main method therefore uses pair-consistent spa-
 508 tial handling. Photometric augmentation can remain independent across slots because it does not
 509 change the geometric relationship between the two observations.

510 **Flow-time and noise sampling.** For the proposed configuration, we use $K = 2$ flow samples per
 511 pair. For each pair i and sample k , we draw $(t_k^{(i)}, \varepsilon_k^{(i)})$ and construct the same noisy action $x_{t_k}^{(i)}$
 512 for both nominal and perturbed views. The same $(t_k^{(i)}, \varepsilon_k^{(i)})$ is shared between the two views of a
 513 matched pair so that the cross-view term compares velocity predictions at the same point on the
 514 action-flow path. The proposed configuration uses $t \sim \text{Beta}(2, 3)$, which concentrates flow-time
 515 samples in the lower-to-mid range of t (mode at $t = \frac{1}{3}$), away from both the pure-noise boundary
 516 $t = 1$ where velocity predictions are uninformative and the near-action boundary $t = 0$ where
 517 predictions have already converged.

518 **Bilateral gradient and shuffled-pair control.** The proposed cross-view loss carries gradients
 519 through both the nominal and perturbed branches. The shuffled-pair mode samples a batch-level
 520 derangement and applies it only inside the cross-view term; all flow-matching labels remain row-
 521 local and correct. This makes the shuffled-pair control a test of action-equivalent pairing rather than
 522 a test of corrupted supervised labels.

523 **Compute.** For the proposed $K = 2$ setting, the vision-language trunk forward depends only on
 524 the view and can be reused across the K flow samples within a pair; only the action expert is re-
 525 evaluated per sample. The per-step wall-clock overhead at $K = 2$ (≈ 2.8 s/step) is therefore below
 526 a naive $2\times$ increase over the single-sample variant (≈ 2.6 s/step). Numbers are measured on a $6\times$
 527 NVIDIA RTX PRO 6000 Server Edition GPU setup.

G Exploratory Feature-Level Alternatives

Before adopting action-flow consistency as the main objective, we explored feature-level alternatives to camera robustness. These experiments are not part of the main evidence for the proposed method, and they are not intended to rule out feature-level methods in general. They explain why the final paper regularizes the policy’s own action-flow field rather than an auxiliary representation.

Canonical-token injection. One approach injected view-stable canonical tokens (output of a trained transformer module from VGGT hidden layer tensors) into the VLA through an additional cross-attention path. This left the dense visual stream available to the action decoder. The corrected canonical-token injection run reached 15.3% camera-track success, comparable to or below the nominal-only floor. Matched, shuffled, and constant canonical-token controls produced similar behavior, suggesting that the policy did not use the sample-specific canonical content as an action-relevant signal.

Canonical residual action anchor. A second approach used canonical tokens to predict a residual action-flow correction gated by a learned scalar. The gate remained nearly closed, and the aggregate camera-track score remained near the nominal-only floor. Forcing the canonical residual branch to be fully active caused nominal-camera rollouts to collapse, indicating that the frozen view-stable representation was not itself an action-executable representation.

Hard action-path bottleneck. A further approach inserted a compact bottleneck between the vision-language trunk and the action decoder and masked dense image-token columns from the action suffix. This made the bottleneck action-grounded, but a same-data attribution test did not support a positive method claim: a standard flow-matching policy trained on the same 30% subset reached 71.4%, while the regularized bottleneck variant reached 67.4%.

Design implication. Across these exploratory attempts, a recurring issue was that the policy could use a high-capacity path around the intended view-stable channel. This motivated the final design in the main paper: instead of requiring a particular hidden feature to become invariant, we directly regularize the action-flow predictions produced by the deployed policy. The matched-vs.-shuffled controls in the main paper then test whether this action-level consistency depends on true action-equivalent pairing.

H Scope Comparison with Related Camera-Aware Methods

Table 9: Inference-contract comparison between representative camera-aware VLA methods and the proposed cross-view action-flow consistency. “Inference-time requirements” lists inputs or computation needed at deployment beyond what our single scene RGB + language + proprioception interface uses.

Family	Representative work	Inference-time requirements	Relation to this work
Camera conditioning	Jiang et al. [11]	Camera extrinsics, Plücker rays, or camera labels supplied to the policy	We use no camera labels or extrinsics at inference
Camera-space action	Zhang et al. [12]	Extrinsic calibration; actions are defined and predicted in the camera frame	We keep actions in the robot-base frame; no calibration is required
Test-time adaptation	Tian et al. [6], Heo et al. [13], Li et al. [19]	A view-synthesis module or a learned representation recalibration applied at deployment	No test-time rendering, synthesis, or representation-adaptation stage
3D / geometric inputs	Qu et al. [7], Abouzeid et al. [8], Li et al. [9, 10], Singh et al. [14]	Depth, point clouds, RGB-D, or learned geometric priors as additional inputs	A single scene RGB image at inference
Multi-view representation	Seo et al. [20], Pang et al. [21]	A view-invariant representation or multi-view world-model encoder	Consistency placed at the action-flow output, not at a hidden representation
This work	—	Single scene RGB, language, proprioception; multi-view pairs are used only during training	—

Scene-camera viewpoint robustness in VLA policies has been approached through several distinct routes, each with its own inference-contract requirements. Table 9 summarizes the principal method families discussed in Section 2 and contrasts them with the proposed objective. The point of this comparison is not to suggest that other families address a different problem, but to make explicit what each family requires at deployment beyond a single scene RGB image, language instruction, and proprioception. Camera-conditioning methods require camera labels or extrinsics; camera-space action formulations require an extrinsic calibration and a non-base action frame; test-time adaptation methods require an additional rendering or representation-recalibration module at deployment; 3D/geometric methods require depth, point clouds, RGB-D, or learned geometric priors; and multi-view representation methods place invariance on a hidden representation or world-model encoder. The proposed method uses none of these at inference, and places consistency directly on the action-flow velocity field that produces rollout behavior. Multi-view pairs are used only during training.

The comparison clarifies why the proposed method is positioned as a training-time recipe rather than as an alternative to camera-aware architectures: it changes the optimization signal during fine-tuning, not the inference interface. Methods in the table that change the inference interface are complementary in principle, and combining them with action-flow consistency is left to future work.

I Future Work

Cross-VLA-family transfer. The proposed objective is stated at the level of an action-flow field v_θ and can in principle apply to any flow- or diffusion-style VLA action head. We evaluate it only in a $\pi_{0.5}$ -style policy. For autoregressive action-token VLAs, the analogous objective would regularize action-token logits or hidden states immediately upstream of the action decoder under the same action-equivalent pairing.

Data-efficient paired fine-tuning. The main simulation experiments use a large same-state paired dataset. In practice, a robot user may want to collect a smaller synchronized multi-camera dataset on a target setup and apply cross-view fine-tuning as a short adaptation stage. A useful follow-up is to sweep the number of action-equivalent pairs and fine-tuning steps required to obtain most of the camera-robustness gain.